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Eric Kleppen Follow

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Dashboards in Python for Beginners using Dash — Responsive Mobile Dashboards with Bootstrap CSS Components

Examples of Bootstrap Components for Responsive Dashboards

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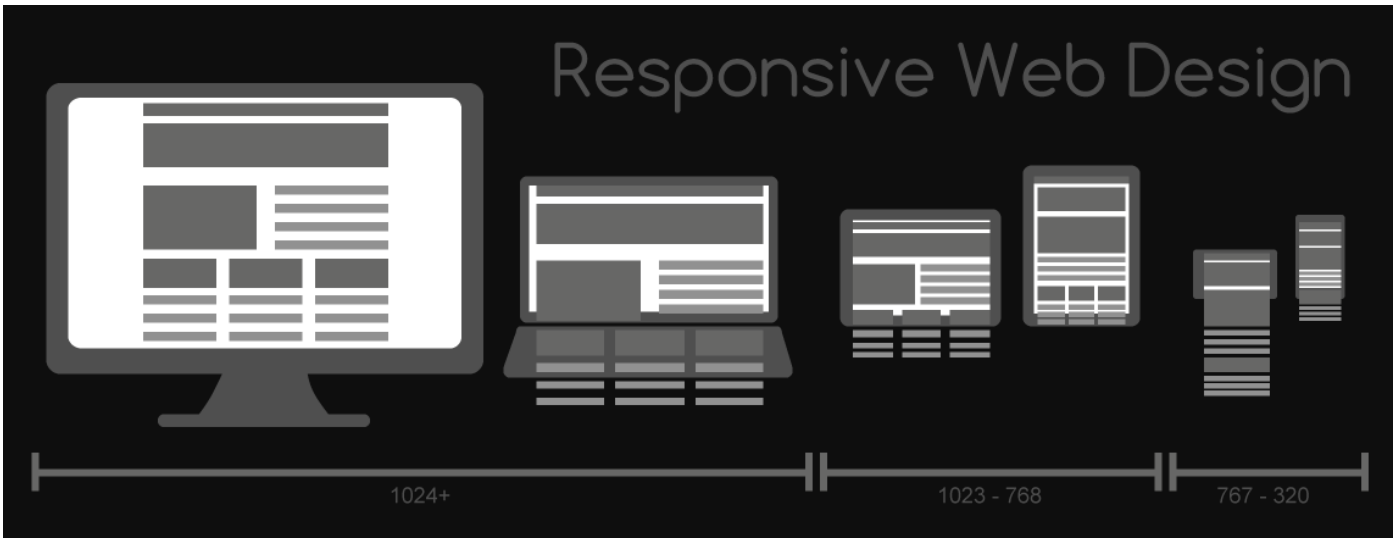
Why Responsive Design

452 3

Mobile browsing is huge, accounting for roughly half of web traffic worldwide, and will likely remain one of the fastest growing ways to consume web media. Since mobile browsing is prevalent, it is important to understand whether the users of your dashboards will need a *responsive design*. For example, at my job, I design dashboards for business people that travel and interact with customers. I need to make sure the visualizations and navigation won't break down if my users need to display the data from their phone or tablet.



https://upload.wikimedia.org/wikipedia/commons/7/7b/Responsive_Web_Design_for_Desktop%2C_Notebook%2C_Tablet_and_Mobile_Phone.png

Although I know enough CSS to style a simple website, I find creating style sheets a little tedious. Dash makes it easy to apply your own style sheets if you're capable of creating them; however, if you're unfamiliar with CSS or don't want to take the time to create your own, Dash can utilize Bootstrap CSS which makes styling and page-layout extremely easy to piece together. Bootstrap is one of the worlds most popular front-end frameworks for creating responsive, mobile friendly sites.

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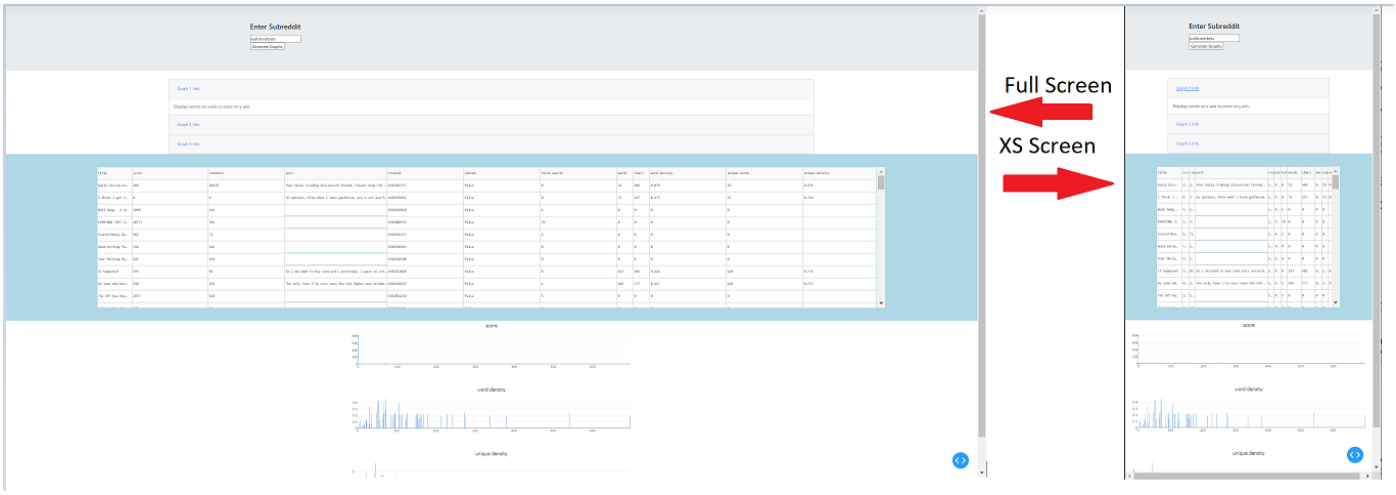


The Completed Dashboard (zoomed out 50% for effect)

Getting into CSS and Dash

In this article, I'll discuss the foundation of the Bootstrap CSS Grid layout, and look at some interesting **dash-bootstrap-components** and *callbacks* like the ***Accordion***, ***Modal*** and ***Jumbotron***.

Dash is a framework for Python written on top of Flask, Plotly.js, and React.js. Dash apps are composed of Layouts and Callbacks:

Layout

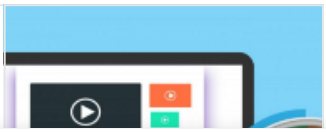
The layout is made up of a tree of components that describe what the application looks like and how users experience the content.

Callbacks

Callbacks make the Dash apps interactive. Callbacks are Python functions that are automatically called whenever an *input* property changes.

If you're unfamiliar with Dash or need a quick refresher, please check out my introduction articles. **All code is available at the end of the article!**

Dashboards in Python for Beginners and Everyone Else using Dash



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Dashboards in Python: 3 Advanced Examples for Dash Beginners and Everyone Else

Code a Python Dashboard! Learn to filter data using dynamic drop lists, create a heatmap, and add a navigation bar...

medium.com

What is CSS?

Cascading style sheets, or CSS, are typically external files that can describe how **HTML** elements in your website are displayed on devices. Style sheets are useful because they can influence multiple pages all at once, thus saving a lot of time and effort if there are several pages in the website. CSS allows me to essentially define the style of my dashboards, and it affects the layout on different screen sizes.

In a traditional CSS file, you would see some syntax like this describing how HTML elements should appear:

example - Notepad

File Edit Format View Help

```
body {
  background-color: blue;
}

H1 {
  color: black;
  text-align: center;
}
H2 {
  color: red;
  text-align: center;
}

P {
  font-family: verdana;
  font-size: 15px;
}
```

Example CSS File

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Bootstrap CSS in Dash

Similar to how the **dash-html-components** library allows you to apply *HTML* using Python, the **dash-bootstrap-components** library allows you to apply Bootstraps front-end components that are affected by the Bootstrap CSS framework. To quote the documentation:

> dash-bootstrap-components *relies on Twitter Bootstrap. To use this package, inject the Bootstrap stylesheet into your application. For convenience, links to Bootstrap CSS hosted on bootstrapcdn are included as part of the* `themes` *module*

To configure the app to run using the Bootstrap CSS, you'll need to include this bit of boilerplate code. Notice this is where the ***external stylesheet*** is set. Since I am using Bootstrap CSS, I set it using **[dbc.themes.BOOTSTRAP]**.

```
app = dash.Dash(__name__, external_stylesheets =
[dbc.themes.BOOTSTRAP])
```

Installation and Dependencies

Install *dash-bootstrap-components* easily using pip or Anaconda:

```
pip install dash-bootstrap-components

OR

conda install -c conda-forge dash-bootstrap-components
```

When importing dependencies, include the **dash_bootstrap_components** library and alias it as **dbc**.

```
import dash
```

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Exploring Bootstrap Layout Components

In my opinion, the Layout Components are the biggest benefit of using the library, but I'll show off a few other cool components after reviewing the grid.

The responsive *grid system* and the convenient *container* wrappers allow for a lot of customization. Bootstrap's grid system uses a series of containers, rows, and columns to layout and align content. Those have been included as components in **dash-bootstrap-components** library as `Container`, `Row`, and `Col`.

> *The `Row` component is a wrapper for columns. The layout of your app should be built as a series of rows of columns.*
>
> *When using the grid layout, content should be placed in columns, and only `Col` components should be immediate children of `Row`.*

The Bootstrap grid contains 12 columns and five tiers of responsiveness. It is easy to visualize the columns using the **dbc.Alert()** component.

```
import dash
import dash_bootstrap_components as dbc
import dash_core_components as dcc
import dash_html_components as html

app = dash.Dash(__name__, external_stylesheets=[dbc.themes.BOOTSTRAP])

body = html.Div([
    html.H1("Bootstrap Grid System Example")
    , dbc.Row(dbc.Col(html.Div(dbc.Alert("This is one column",
color="primary"))))
    , dbc.Row([
            dbc.Col(html.Div(dbc.Alert("One of three columns",
color="primary")))
            , dbc.Col(html.Div(dbc.Alert("One of three columns",
color="primary")))
            , dbc.Col(html.Div(dbc.Alert("One of three columns",
```

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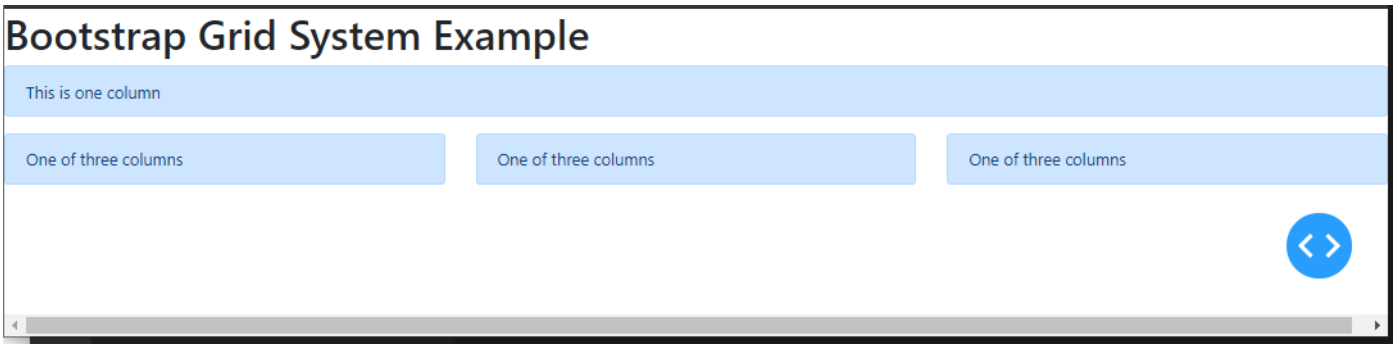
```
if __name__ == "__main__":
    app.run_server(debug = True)
```

Notice the body is built using a series of *Rows* and *Columns* like the documentation suggests. Notice **dbc.Col** is the immediate child of **dbc.Row**, but **dbc.Col** is followed by familiar Dash *HTML* components.



By default, the column will expand to take the entire screen. To have more control over how the columns expand, try wrapping the grid in a **dbc.Container** or use the **width** argument to assign a *width* to the column.

Notice the columns have a *gap* in between them. The spaces are called *Gutters* and are a part of the **dbc.Row** component. To remove the gutters set **no_gutters = True**.

Below is a more extensive example of how the grid can be styled. Since the rows make up 12 columns, it is easiest to use numbers that factor 12 when working with multiple columns in a row. As seen in the example, I use several combinations of column sizes.

```
body = html.Div([html.H1("Bootstrap Grid System Example")
, html.H4("no_gutters = False")
, dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3)
])
```

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```
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3)
], no_gutters = True)
, html.H3("Examples of justify property")
, html.H4("start, center, end, between, around")
, dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=4),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=4),
],
justify="start")
, dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
],
justify="center")
, dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of three columns")), width=3)
    , dbc.Col(html.Div(dbc.Alert("One of three columns")), width=3)
    , dbc.Col(html.Div(dbc.Alert("One of three columns")), width=3)
],
justify="end")
, dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3)
    , dbc.Col(html.Div(dbc.Alert("One of three columns")), width=3)
],
justify="between")
, dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=4),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=4),
],
justify="around")
, html.H4("Container Example")
, dbc.Container([
    dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3)
])

, html.H4("no_gutters = True")
, dbc.Row([
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3),
    dbc.Col(html.Div(dbc.Alert("One of two columns")), width=3)
```

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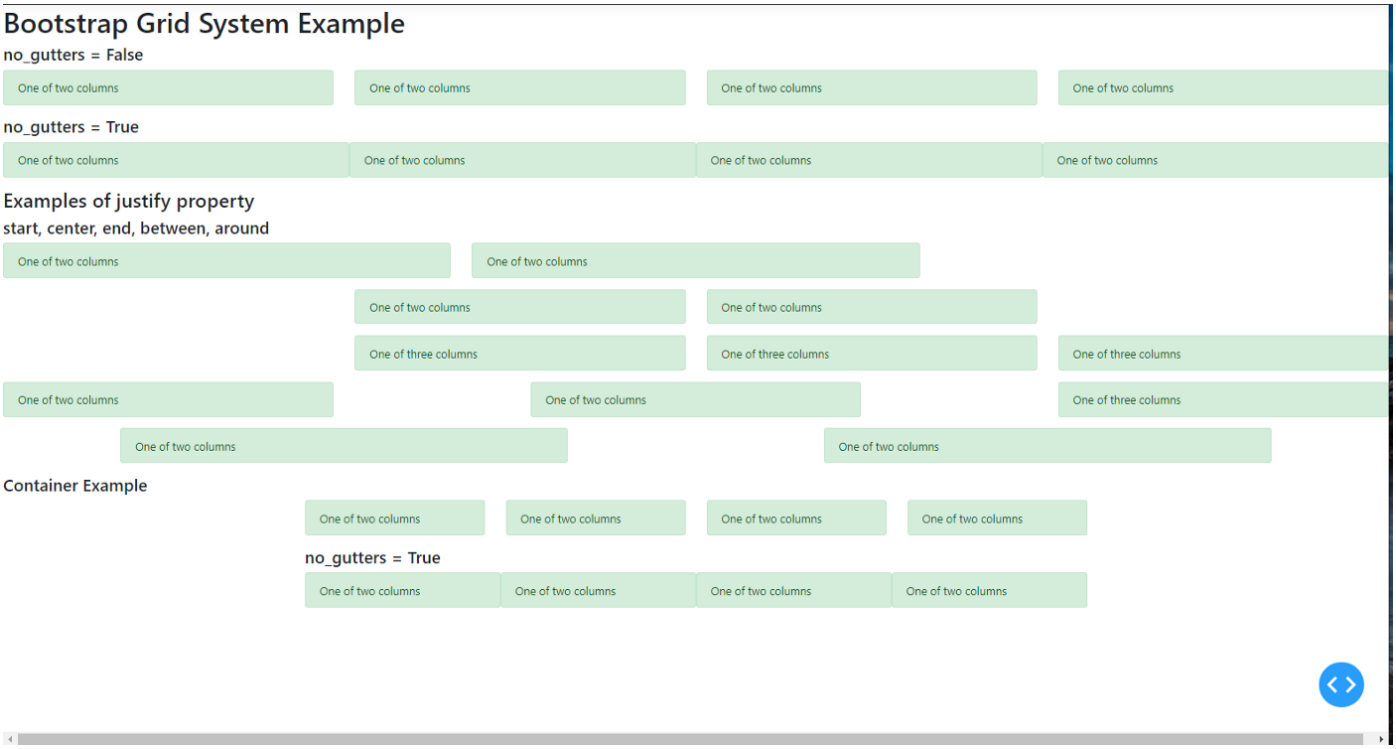

Bootstrap Grid System

Notice the **gutters** can be easily removed, and **Rows** can be *justified*.
Notice how the *container example* centers and shrinks the columns by default.

In addition to using the *width* argument, customize the size, order and offset of your **columns** on a variety of devices using the `xs`, `sm`, `md`, `lg`, and `xl` keyword arguments.

```
body = html.Div([
html.H1("Bootstrap Grid System Example")
, dbc.Row(dbc.Col(html.Div(dbc.Alert("This is one column",
color="primary"))))
, dbc.Row([
        dbc.Col(html.Div(dbc.Alert("One of three columns",
color="primary")), lg=3, md=4, xs=12)
        , dbc.Col(html.Div(dbc.Alert("One of three columns",
color="primary")), lg=3, md=4, xs=12)
        , dbc.Col(html.Div(dbc.Alert("One of three columns",
color="primary")), lg=3, md=4, xs=12)
        ])
    ])
```

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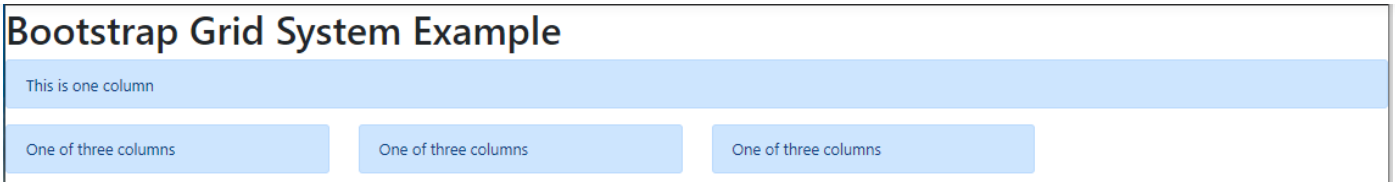
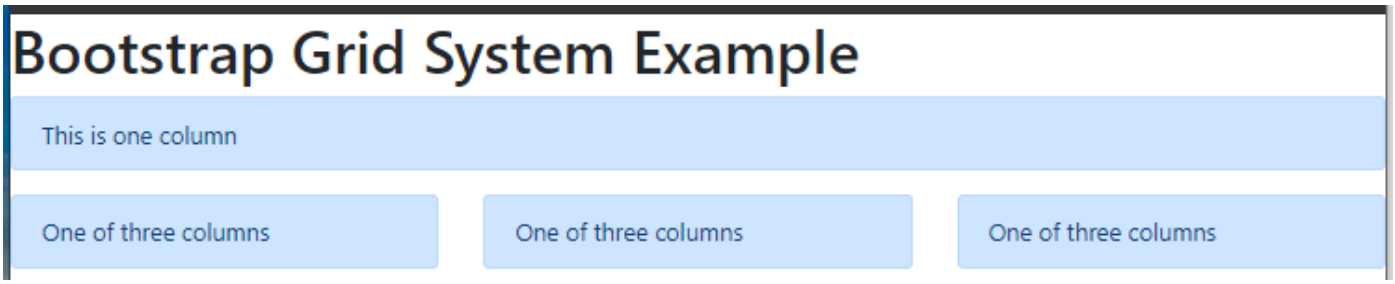
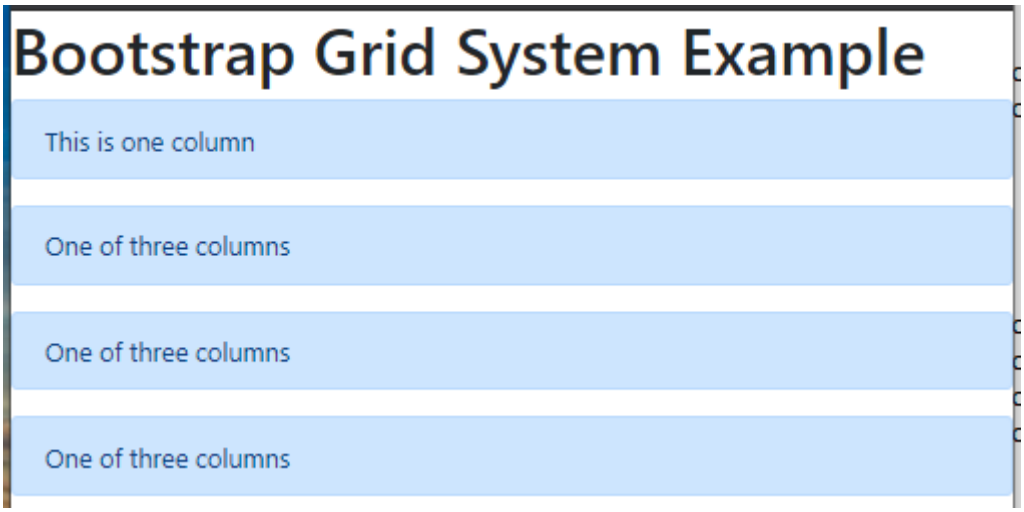


Large (lg = 3)



Medium (md = 4)



Extra Small (xs = 12)

Notice the column size changes as the screens get smaller. For example, using **xs = 12** causes the column to consume the entire row when the screen is extra small since a the grid system is composed of 12 columns.

That, in a nutshell, is the simplicity and power of the Bootstrap responsive grid system. In a few lines of code, it is possible to create a responsive layout that will work on a variety of screen sizes. It is easy to apply the grid to existing dashboards.

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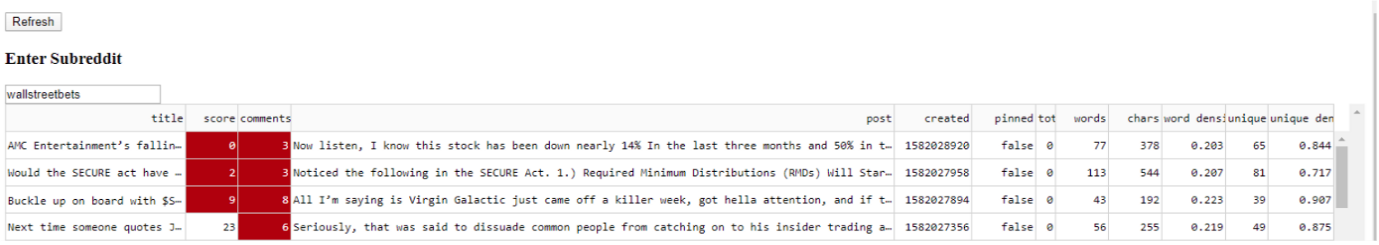
components, I'll apply them to the Reddit dashboard. **If you're already familiar with Reddit, I include all of the starter code below in this article**; otherwise, refer to the previous article for instructions on generating Reddit API keys!



title	score	comments	post	created	pinned	tot	words	chars	word dens	unique	unique den
AMC Entertainment's fallin…	0	3	Now listen, I know this stock has been down nearly 14% In the last three months and 50% in t…	1582028920	false	0	77	378	0.203	65	0.844
Would the SECURE act have …	2	3	Noticed the following in the SECURE Act. 1.) Required Minimum Distributions (RMDs) Will Star…	1582027958	false	0	113	544	0.207	81	0.717
Buckle up on board with $S…	9	8	All I'm saying is Virgin Galactic just came off a killer week, got hella attention, and if t…	1582027894	false	0	43	192	0.223	39	0.907
Next time someone quotes J…	23	6	Seriously, that was said to dissuade common people from catching on to his insider trading a…	1582027356	false	0	56	255	0.219	49	0.875

The Reddit Dashboard

Dashboards in Python Using Dash — Creating a Data Table using Data from Reddit

Mine Reddit for NLP and Share with Dash. Create a dashboard that displays posts from an entered subreddit. Display NLP…

medium.com

I'll explain how I turn the original Reddit Dashboard into the image below using Bootstrap CSS. I'll add a *Jumbotron*, a *Collapse Accordion*, and a *Modal* that generates a popup when the **Generate Graphs** button is pushed:

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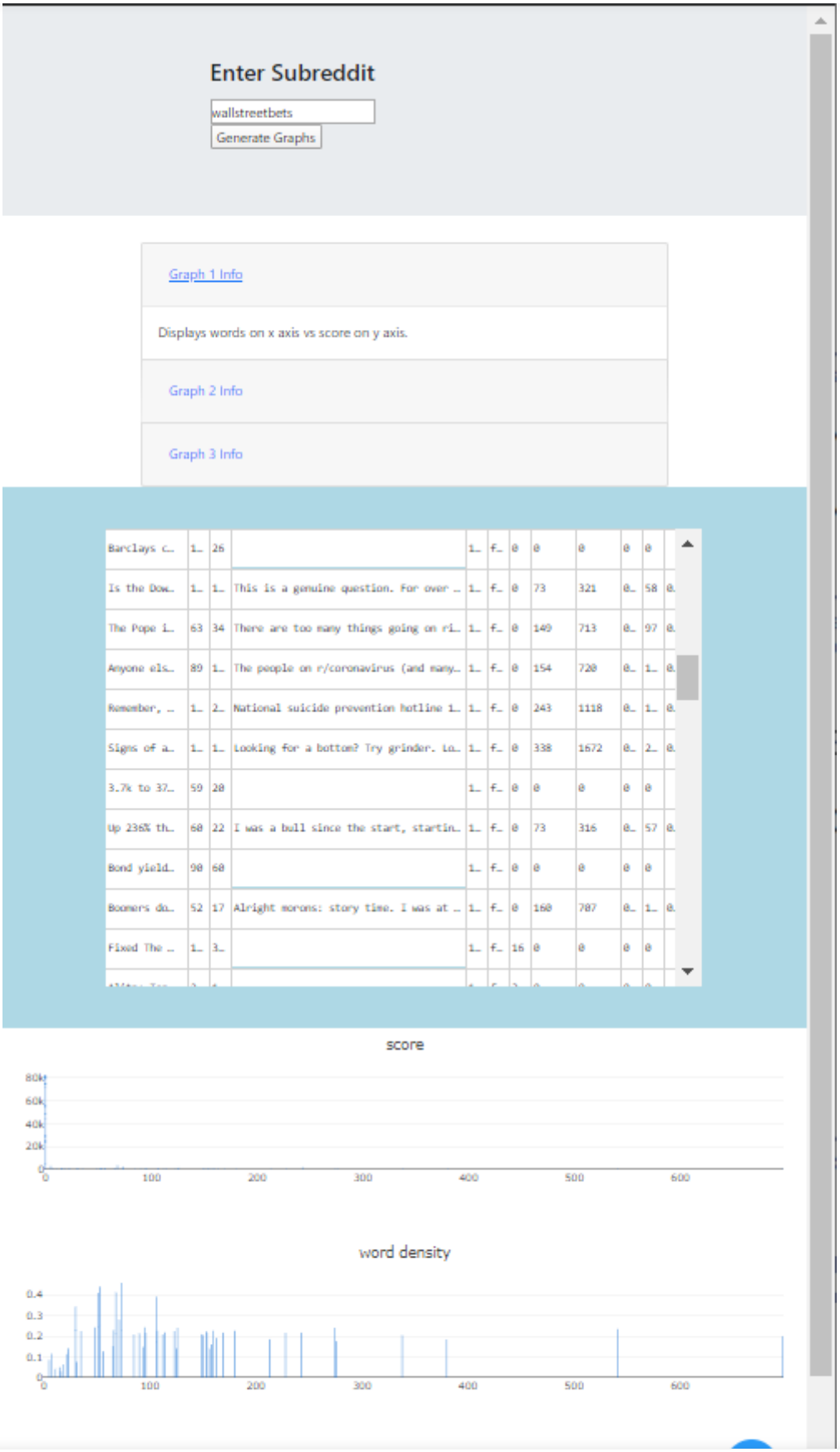

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Starter Code

This is starter code that will provide you with the core Reddit dashboard. It assumes you are using a **config** file to pass the API keys into the code. It defaults to the *subreddit* **wallstreetbets**, so update that value if you'd like it to default to a different one.

```
import dash
import dash_html_components as html
import dash_core_components as dcc
import dash_table
import pandas as pd
import praw
import pandas as pd
from dash.dependencies import Input, Output, State
from dash.exceptions import PreventUpdate
from config import cid, csec, uag

reddit = praw.Reddit(client_id= cid, client_secret= csec, user_agent=
uag)

posts = []

new_bets = reddit.subreddit('wallstreetbets').new(limit=100)

for post in new_bets:
    posts.append([post.title, post.score, post.num_comments,
post.selftext, post.created, post.pinned, post.total_awards_received])
posts = pd.DataFrame(posts,columns=['title', 'score', 'comments',
'post', 'created', 'pinned', 'total awards'])

def get_features(dataframe):
    df = posts.copy()

df['words'] = df['post'].apply(lambda x : len(x.split()))
    df['chars'] = df['post'].apply(lambda x : len(x.replace(" ","")))
    df['word density'] = (df['words'] / (df['chars'] + 1)).round(3)
    df['unique words'] = df['post'].apply(lambda x: len(set(w for w in
x.split())))
    df['unique density'] = (df['unique words'] / df['words']).round(3)

return df

df = get_features(posts)
```

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```
value='wallstreetbets')
    ,dash_table.DataTable(
    id='table'
    ,columns=[{"name": i, "id": i} for i in df.columns]
    ,fixed_rows={ 'headers': True, 'data': 0 }
    ,data=df.to_dict('records')
)
])

@app.callback(Output('table', 'data'),
              [Input('refresh', 'n_clicks')],
              [State('input-1-state', 'value')
               ])
def update_data(n_clicks, subreddits):
    dff = df
    if subreddits is None:
        subreddits = 'wallstreetbets'
    else:
        subreddits
    if n_clicks is None:
        raise PreventUpdate
    else:
        posts = []
        new_bets = reddit.subreddit(subreddits).hot(limit=100)
        for post in new_bets:
            posts.append([post.title, post.score, post.num_comments,
post.selftext, post.created, post.pinned, post.total_awards_received])
        posts = pd.DataFrame(posts,columns=['title', 'score',
'comments', 'post', 'created', 'pinned', 'total awards'])
        dff = get_features(posts)

    return dff.to_dict('records')
```

Adding Jumbotron

A Jumbotron is a lightweight component that helps content stand out. It takes up a lot of space on the screen which makes it great for showcasing buttons and messages. **The Jumbotron is this big gray box surrounding the input field and button**.

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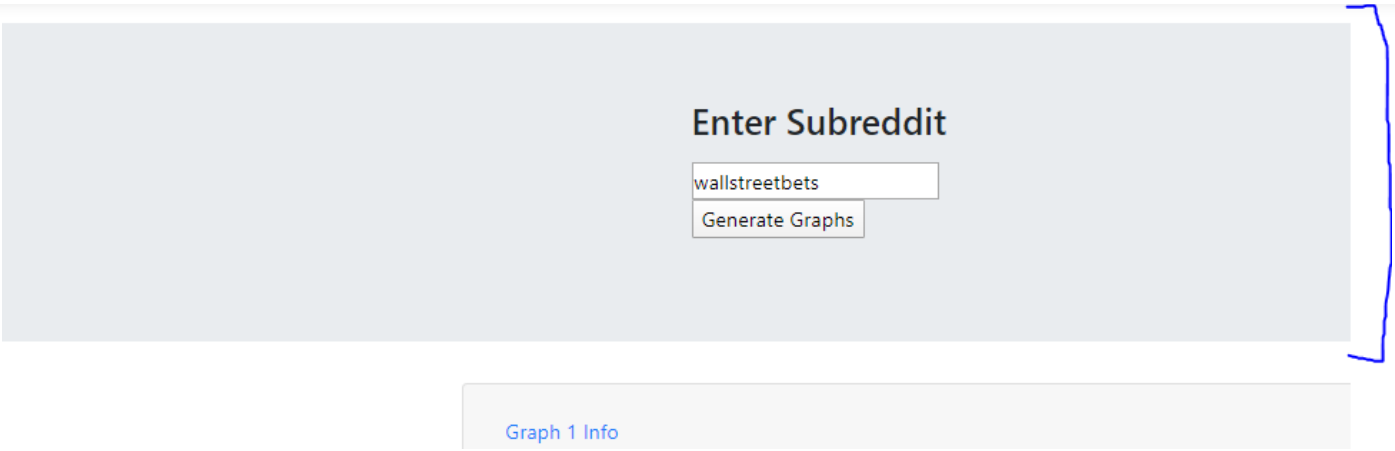

The Jumbotron

Here is the section of code being adjusted for the Bootstrap component:

```
# Original
html.Div([
    html.P(html.Button('Refresh', id='refresh'))
    ,html.P(html.Div(html.H3('Enter Subreddit')))
    ,dcc.Input(id='input-1-state', type='text',
value='wallstreetbets')
])
```

I will use **dbc.Jumbotron** to wrap the **html.Div** that contains the input and button components. The bold code is what has wrapped the original code.

```
dbc.Jumbotron([
dbc.Row(dbc.Col([
html.P(html.Div(html.H3('Enter Subreddit')))
, dcc.Input(id='input-1-state', type='text', value='wallstreetbets')
, html.P(html.Button('Generate Graphs', id='refresh'))
   ], width = 6)
  , justify = 'center')# end row
  ], fluid = True)
```

Notice the **dbc.Jumbotron** wraps **dbc.Row** and **dbc.Col** so I can take advantage of the

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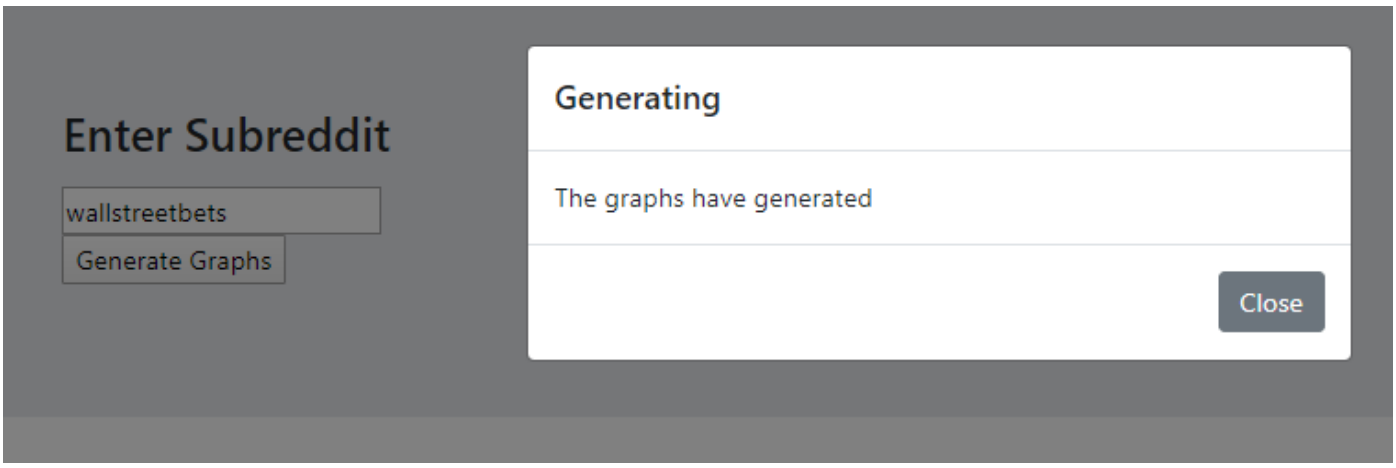
Adding Modal

Modals are another useful tool for drawing attention to messages or features. Use the **dbc.Modal component** to add user notifications or custom content that pops up after the user performs an action. For example, when a user clicks the *Generate Graphs* button, Modal produces a popup that lets the user know the graphs have been generated.



Modal Example

The Modal is being placed within the Jumbotron. I put it there since it interacts with the button.

```
dbc.Jumbotron([
dbc.Row(dbc.Col([
html.P(html.Div(html.H3('Enter Subreddit')))
, dcc.Input(id='input-1-state', type='text', value='wallstreetbets')
, html.P(html.Button('Generate Graphs', id='refresh'))
, dbc.Modal(
     [
   dbc.ModalHeader("Generating"),
   dbc.ModalBody("The graphs have generated"),
   dbc.ModalFooter(
      dbc.Button("Close", id="close", className="ml-auto")
      ),
      ],
      id="modal",
    )
   ], width = 6)
```

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Notice the *Modal* is constructed using a **Header**, **Body** and **Footer** component. By default, the Modal can be dismissed by clicking outside the modal or by pressing the escape key.

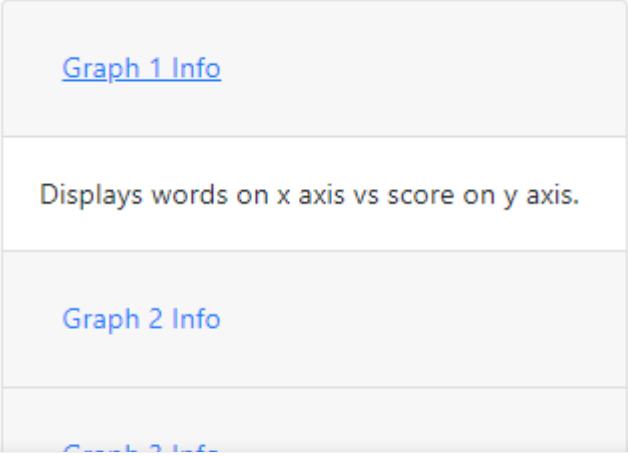
It is possible to adjust the size of the Modal using the **size** property. The options are `sm`, `lg`, or `xl` for a small, large or extra large Modal.

In order to function, the Modal uses a *callback*. Using callbacks, it is possible to set the **is_open** property to **True** or **False**, depending on how I want it to display.

```
@app.callback(
    Output("modal", "is_open"),
    [Input("open", "n_clicks"), Input("close", "n_clicks")],
    [State("modal", "is_open")],
)
def toggle_modal(n1, n2, is_open):
    if n1 or n2:
        return not is_open
    return is_open
```

Adding Accordion with Collapse

By using the **dbc.Collapse component**, it is possible to toggle the visibility of content. By combining the **Collapse** and the **Card** component, you can extend the default collapse behavior to create an accordion.



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Creating an Accordion is tricky because Dash will throw an error if components have the same ID. Thus, using a function, I can programmatically create unique IDs for the **Collapse** and **Card** components in the accordion. This also makes populating the Cards in the accordion a bit difficult, but I use a list in the function to simplify the population logic!

```
def make_item(i):
    # we use this function to make the example items to avoid code
duplication
    graph_info = [0,'Displays words on x axis vs score on y axis.'
                  , 'Displays words on x axis vs word density on y
axis.'
                  , 'Displays words on x axis vs unique density on y
axis.']
    return dbc.Card(
        [
            dbc.CardHeader(
                html.H2(
                    dbc.Button(
                        f"Graph {i} Info",
                        color="link",
                        id=f"group-{i}-toggle",
                    )
                )
            ),
            dbc.Collapse(
                dbc.CardBody(graph_info[i]),
                id=f"collapse-{i}",
            ),
        ]
    )
```

The function ***make_item(i)*** takes in a numeric value and outputs a **dbc.Card** component that wraps the *Collapse* used to create the accordion effect.
Notice the list **graph_info.** It is the list I use to populate the contents of the **dbc.CardBody** since I can pass the corresponding numeric value to the index of the list. For example, if I pass 1 into **make_item(1)**, I will return the value at index 1 in my list **graph_info**. The item in index one is this:

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Constructing the Accordion

To construct the accordion, I call the functions and wrap them in an html.Div component.

```
accordion = html.Div([make_item(1), make_item(2), make_item(3)],
className="accordion")
```

I add it to the main html.Div list between the jumbotron and the data table.

Accordion Callback

I took the boilerplate callback from the documentation. Depending on the number of cards in the accordion, increase the range for the **Output**, **Input** and **State**.

```
@app.callback(
    [Output(f"collapse-{i}", "is_open") for i in range(1, 4)],
    [Input(f"group-{i}-toggle", "n_clicks") for i in range(1, 4)],
    [State(f"collapse-{i}", "is_open") for i in range(1, 4)],
)
def toggle_accordion(n1, n2, n3, is_open1, is_open2, is_open3):
    ctx = dash.callback_context

if not ctx.triggered:
        return ""
    else:
        button_id = ctx.triggered[0]["prop_id"].split(".")[0]

if button_id == "group-1-toggle" and n1:
        return not is_open1, False, False
    elif button_id == "group-2-toggle" and n2:
        return False, not is_open2, False
    elif button_id == "group-3-toggle" and n3:
        return False, False, not is_open3
    return False, False, False
```

The Complete Dashboard

After completing the dashboard, it is easy to see how Bootstrap CSS for Dash is a powerful

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a dashboard's usability. Below is the complete code for the dashboard. Check out my other tutorials if you're new to coding or data science too!

Dashboards in Python for Beginners using Dash — Live Updates and Streaming Data into a Dashboard

Add a Live Feed to Dash. This tutorial shows how simple it is to pull data out of a database and display it through a…

levelup.gitconnected.com

Simple Sentiment Analysis for NLP Beginners and Everyone Else using VADER and TextBlob

Extract sentiment metrics from text in a Panda's DataFrame in a few lines of code without training models!

medium.com

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— Eric Kleppen

The Code

Here is the complete code. Remember, it assumes you are using a config file to import the API keys.

```
import dash
```

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```
import pandas as pd
from dash.dependencies import Input, Output, State
from dash.exceptions import PreventUpdate
from config import cid, csec, uag

reddit = praw.Reddit(client_id= cid, client_secret= csec, user_agent=
uag)

posts = []

new_bets = reddit.subreddit('wallstreetbets').hot(limit=100)

for post in new_bets:
    posts.append([post.title, post.score
    , post.num_comments, post.selftext
    , post.created, post.pinned, post.total_awards_received])
posts = pd.DataFrame(posts,columns=['title', 'score', 'comments',
'post', 'created', 'pinned', 'total awards'])

def get_features(dataframe):
    df = posts.copy()

df['words'] = df['post'].apply(lambda x : len(x.split()))
    df['chars'] = df['post'].apply(lambda x : len(x.replace(" ","")))
    df['word density'] = (df['words'] / (df['chars'] + 1)).round(3)
    df['unique words'] = df['post'].apply(lambda x: len(set(w for w in
x.split())))
    df['unique density'] = (df['unique words'] / df['words']).round(3)

return df

df = get_features(posts)

def make_item(i):
    # we use this function to make the example items to avoid code
duplication
    graph_info = [0,'Displays words on x axis vs score on y axis.'
                    , 'Displays words on x axis vs word density on y
axis.'
                    , 'Displays words on x axis vs unique density on y
axis.']
    return dbc.Card(
        [
            dbc.CardHeader(
                html.H2(
```

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```
                )
            ),
            dbc.Collapse(
                dbc.CardBody(graph_info[i]),
                id=f"collapse-{i}",
            ),
        ]
    )

accordion = html.Div(
    [make_item(1), make_item(2), make_item(3)], className="accordion"
)

app = dash.Dash(__name__, external_stylesheets=[dbc.themes.BOOTSTRAP])

app.layout =  html.Div([

dbc.Jumbotron([

dbc.Row(dbc.Col([
        html.P(html.Div(html.H3('Enter Subreddit')))
        , dcc.Input(id='input-1-state', type='text',
value='wallstreetbets')
        , html.P(html.Button('Generate Graphs', id='refresh'))
        , dbc.Modal(
                [
                    dbc.ModalHeader("Generating"),
                    dbc.ModalBody("The graphs have generated"),
                    dbc.ModalFooter(
                        dbc.Button("Close", id="close", className="ml-
auto")
                    ),
                ],
                id="modal",
            )
        ], width = 6)
        , justify = 'center')# end row
        ], fluid = True)

    , html.Div(dbc.Row(dbc.Col(accordion ,width = 8),justify =
'center'))

    , html.Div([
        dbc.Row(dbc.Col(dash_table.DataTable(
        id='table'
        columns=[{"name": i, "id": i} for i in df.columns]
```

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```
            ,'padding': '15px'},
            {'if': {'column_id': 'post'},
            'width': '400px'
            }
            ,{'if': {'column_id': 'words'},
            'width': '65px'
            }
            ,{'if': {'column_id': 'chars'},
            'width': '65px'
            }
        ]
        ,style_cell={
            'overflowX': 'hidden',
            'textOverflow': 'ellipsis',
            'maxWidth': '25px',
        'textAlign': 'left'
        }
        , style_table={
            'maxHeight': '550px'
            #,'maxWidth': '800px'
            ,'overflowY': 'scroll'
            ,'overflowX': 'hidden'
        }

), width = 10), justify = 'center')# end dt
    ], style = {'background-color': 'lightblue'
                #,'margin-bottom': '5px'
                , 'padding' : '50px'
                    }) #end div

,  dbc.Container(
            html.Div(id = 'graph-container')
    )

])#end div

@app.callback(
    [Output(f"collapse-{i}", "is_open") for i in range(1, 4)],
    [Input(f"group-{i}-toggle", "n_clicks") for i in range(1, 4)],
    [State(f"collapse-{i}", "is_open") for i in range(1, 4)],
)
def toggle_accordion(n1, n2, n3, is_open1, is_open2, is_open3):
    ctx = dash.callback_context

if not ctx.triggered:
        return ""
```

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```
    elif button_id == "group-2-toggle" and n2:
        return False, not is_open2, False
    elif button_id == "group-3-toggle" and n3:
        return False, False, not is_open3
    return False, False, False

@app.callback(
    Output("modal", "is_open"),
    [Input("refresh", "n_clicks"), Input("close", "n_clicks")],
    [State("modal", "is_open")],
)
def toggle_modal(n1, n2, is_open):
    if n1 or n2:
        return not is_open
    return is_open

@app.callback(
    Output('graph-container', "children"),
    [Input('table', "data")])
def update_graph(rows):
    dff = pd.DataFrame(rows)
    return html.Div(
        [
            dcc.Graph(
                id=column,
                figure={
                    "data": [
                        {
                            "x": dff["words"],
                            "y": dff[column] if column in dff else [],
                            "type": "bar",
                            "marker": {"color": "#0074D9"},
                        }
                    ],
                    "layout": {
                        "xaxis": {"automargin": True},
                        "yaxis": {"automargin": True},
                        "height": 250,
                        "margin": {"t": 50, "l": 10, "r": 10},
                        "title": column
                    },
                },
            )
            for column in ["score", "word density", "unique density"]
        ]
    )
```

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```
def update_data(n_clicks, subreddits):
    dff = df
    if subreddits is None:
        subreddits = 'wallstreetbets'
    else:
        subreddits
    if n_clicks is None:
        raise PreventUpdate
    else:
        posts = []
        new_bets = reddit.subreddit(subreddits).hot(limit=100)
        for post in new_bets:
            posts.append([post.title, post.score, post.num_comments,
post.selftext, post.created, post.pinned, post.total_awards_received])
        posts = pd.DataFrame(posts,columns=['title', 'score',
'comments', 'post', 'created', 'pinned', 'total awards'])
        dff = get_features(posts)

    return dff.to_dict('records')

if __name__ == '__main__':
    app.run_server(debug=True)
```

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